



London Fire Brigade C1 analysis of vehicle arrival times

Group Members: Matias Muñoz Hoffmann and Clemente Ibarra

Course: Business Intelligence

Institution: Universidad Del Desarrollo

Professor: Tomás Fontecilla Correa

Submission date: 29/09/2026

Table of Contents

Problem, foundation and evolution from W1	3
Seven step map from Class 03	3
Sources, preparation and quality	3
Five Cs	4
Exploratory findings and interpretation	5
Conclusion and feasible continuation	6
Process record	7
Decisions and checks	7
Individual contributions	7
AI use and verification.....	8
Individual reflections confirmed by team representative	8
References.....	8

Problem, foundation and evolution from W1

A proposed London Fire Brigade (LFB) operations planning analyst could use a descriptive shortlist to decide which boroughs and dispatch hours warrant a closer review of long vehicle mobilization to arrival times. This is a proposed user and decision, not a statement about LFB's current workflow. The question is: which boroughs and dispatch hours have longer observed times in the published 2023-2024 mobilisation records, and where is the pattern persistent? The analysis does not recommend changes to staffing, stations or dispatch rules.

The W1 exercise compared two lines using the same public dataset: (A) geographic differences in attendance time and (B) temporal variation in the number of mobilisations. It provisionally preferred A because the duration field and borough label support a direct, bounded comparison with LFB's response-time interest. C1 retains A, narrows the window to dispatches in 2023-2024, uses P90 alongside median and record count, and adds an hour of dispatch view as a supporting comparison. No instructor feedback on that choice has been supplied to this team record. The W1 choice therefore remains provisional pending feedback. The W1 report is included as data/W1_selection_evidence.pdf.

Official context is LFB's Community Risk Management Plan 2023-2029 and its published attendance-time methodology. They establish relevance, not an official numerical target for this C1 KPI. The present measure includes all eligible published vehicle arrivals, not only the first vehicle at an incident.

Seven step map from Class 03

Step	C1 application
1. Decision	Proposed user: LFB operations planning analyst. Decision: prioritise deeper investigation by borough and dispatch hour before any resource change.
2. Success criteria	Desired outcome: more reliable and equitable attendance. Current data can describe distributions and identify candidates, it cannot establish an improvement from an intervention.
3. Business questions	Main: where are longer observed mobilisation-to-arrival times? Supporting: how do median, P90, volume and persistence vary across geography and hour?
4. Business process	A call precedes mobilization, a vehicle is mobilised, becomes mobile and arrives. This file measures mobilisation to arrival, it excludes call handling and does not identify all operational conditions.
5. Entities, events and states	One row is one vehicle mobilisation. IncidentNumber links multiple vehicles to one incident. ResourceMobilisationId identifies a mobilisation. Published rows are labelled Initial.
6. Data representation	Twenty four columns in the original CSV. One row per published mobilisation. No external join is used, because a validated common key and compatible unit are not available for this question.
7. Measurement	Attendance minutes = AttendanceTimeSeconds / 60. P90 is the 0.9 quantile of eligible row values (linear interpolation). Report n, median and P90 globally and by group. Denominator for shares is eligible mobilisations in the same scope, missing days are not zeros.

Sources, preparation and quality

The numerical source is the LFB mobilisation CSV for 2021-2024 and its metadata dictionary, retrieved 23 September 2026. data/source_manifest.json records the public URLs, size and SHA256. The exact original is preserved in gzip form in data/raw;

its uncompressed SHA256 is checked by analysis/analysis.py. data/reduced contains an uncleaned 2023-2024 extract generated by selecting the year in DateAndTimeMobilised, without deduplication or value changes. Prepared tables and figures are in data/case_tables. The executed notebook and analysis code trace the complete path from the original to those results.

Stage	Rows
Source 2021-2024	727,857
After exact deduplication	723,935
Selected dispatch dates 2023-2024	384,643
KPI eligible	384,627

The first 3,922 removed rows are exact copies after whitespace, date and number normalisation. Selecting the dispatch window then leaves 384,643 rows. Sixteen period rows have conflicting mobilisation IDs and are excluded because choosing one location would be arbitrary. The final 384,627 rows correspond to 248,225 distinct incident IDs. The full source profile, conflicting-ID table, exclusion table and row flow table are supplied. The notebook checks identifier uniqueness, nonnegative durations and agreement between recorded duration and arrival minus mobilisation (tolerance one second). One valid zero duration remains. Missing geography (1,288 eligible rows) stays as Unknown in the global KPI, it is not assigned a borough. Missing turnout/travel components do not invalidate a coherent total, but component correlations use only complete, consistent rows (n=382,811). No target imputation or arbitrary removal of long valid times is applied.

Five Cs

Lens	Check and treatment
Clean	Profile types, missingness, exact duplicates, IDs, timestamp consistency and negative durations. Remove exact copies and quarantine 16 conflicting ID period rows, retain valid zero and long times.
Consistent	Use the same eligibility and P90 calculation in both years. Date parsing, units and timestamp basis are fixed. The absent 31 December 2024 prevents a complete calendar year volume comparison.
Conformed	No cross source join is used. Original dictionary and CSV field names were checked, the older dictionary omits geography labels, which are treated as published source labels rather than external reference codes.
Current	The 2021-2024 publication supports a historical 2023-2024 diagnostic, not a real time 2026 dispatch decision. Any operational deployment would require updated records and context.
Comprehensive	Published pumping appliance records, not every actual emergency. No 31 December 2024 rows, no durations above 20 minutes, and all observed rows labelled Initial. Incident mix, distance, traffic and resource availability are not controlled.

No join is forced. Joining this file to a population table by borough would not make population a valid denominator for vehicle durations. The proposed questions use the observed mobilisation unit directly. Source methodology discusses exclusions in some official performance calculations, including times above 20 minutes; this is consistent with, but does not prove, every selection rule for this CSV.

Exploratory findings and interpretation

For 384,627 eligible published mobilisations, median attendance is 5.65 minutes, mean 6.03, and P90 9.20. P90 means roughly 90% of these observed eligible vehicle times are at or below 9.20 minutes. This is not the share of first appliance incidents meeting an official target.

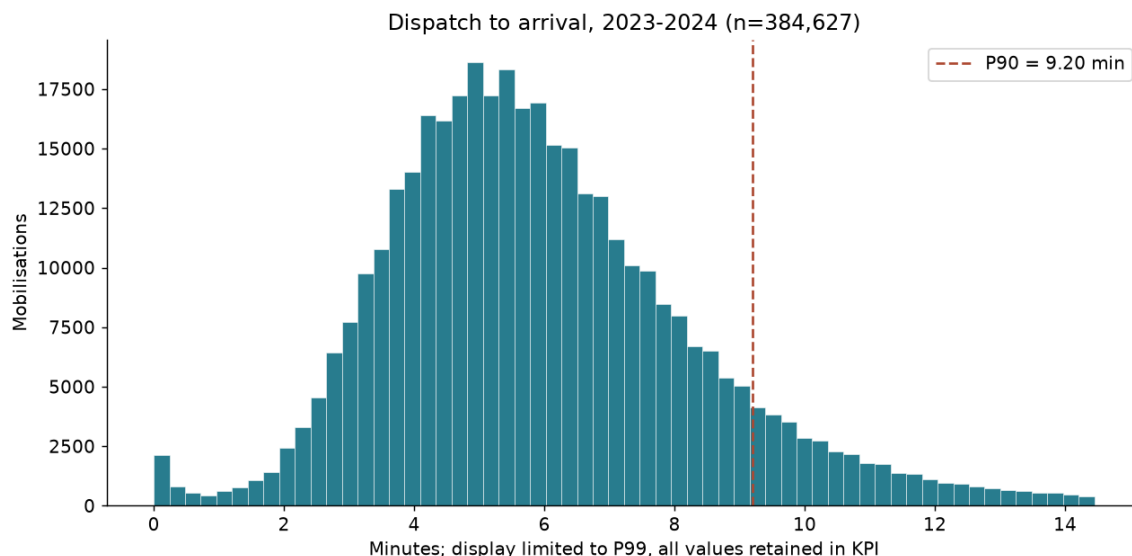


Figure 1. Attendance-time distribution to P99 for display. All eligible values remain in the KPI.

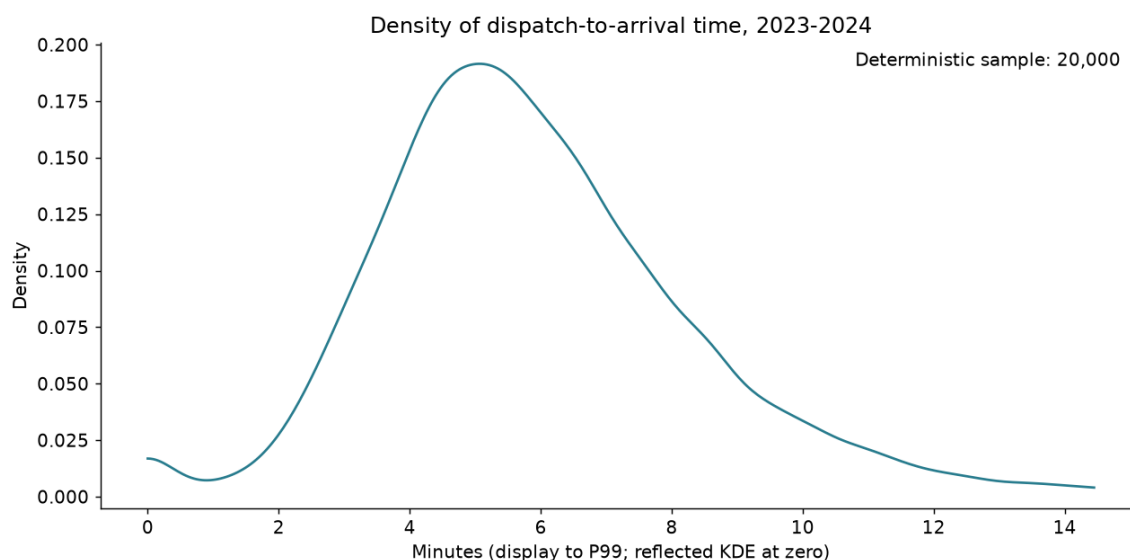


Figure 2. Density from a reproducible 20,000-row sample; curve height is not a direct percentage.

Borough	Eligible n	Median min	P90 min
Hillingdon	12,195	6.63	10.67
Havering	7,033	6.25	10.22
Enfield	10,178	6.20	10.15
Bromley	9,848	6.25	10.10
Redbridge	7,824	6.10	9.95

Hillingdon has the highest observed borough P90 (10.67 minutes; 12,195 eligible mobilisations). This supports prioritising contextual review, not a finding of poor crew performance. Distance, incident mix and deployment conditions are plausible alternative explanations that the extract does not resolve.

The highest dispatch hour P90 is 10.03 minutes for 11:00 - 11:59 GMT (n=19,425). This pools all available dates and does not establish traffic as the cause or imply higher volume. Spearman's correlation of travel with total attendance is 0.952 on 382,811 complete, consistent component records, total includes travel by construction, so this coefficient is not a percentage of total time or a causal effect.

Dispatch year	Eligible n	Median min	P90 min
2023	186,339	5.62	9.17
2024	198,288	5.68	9.25

The year to year P90 difference is small and descriptive. It cannot establish deterioration because 2024 lacks its last day and the case mix may differ. The source's 20 minute observed ceiling also prevents claims about the missing extreme tail. Missing records are distinct from recorded zero durations.

An exploratory review rule flags a borough whose monthly P90 exceeds the global P90 for the same month, with at least 100 eligible records and at least three consecutive qualifying months. Excluding incomplete December 2024, 19 boroughs meet the three month condition, 15 meet six months. Those counts are unchanged when the minimum n is 50, 100 or 200 in the saved sensitivity table. This is not an official LFB threshold or a statistically validated trigger.

Instructor feedback and response

The instructor did not provide feedback on our W1 project selection. We therefore retained the geographic analysis of London Fire Brigade mobilisation to arrival times as our working direction for C1. We narrowed the scope to 2023 - 2024 and developed the data-quality checks, exploratory analysis and comparisons described in this report. These are team decisions, not instructor approved requirements or official LFB performance targets. If feedback is provided later, we will review the scope and methodological choices before continuing with C2.

Conclusion and feasible continuation

The bounded answer is that borough and dispatch-hour distributions differ in the published 2023-2024 population. Hillingdon and the 11:00 GMT group are examples for review. Before acting, would need updated coverage, incident type, route distance, vehicle availability and local operational context. No causal or service quality ranking follows from these descriptive records.

Possible ML task for C2: predict mobilisation-to-arrival seconds for a newly mobilised vehicle, using only fields actually available at dispatch, such as dispatch calendar information and validated origin/destination labels. The unit remains one mobilisation, the supervised outcome is the recorded duration. Actual travel time, turnout time, arrival timestamp and delay reason must not be predictors at dispatch. Feasibility depends on field availability, a leakage audit and later date validation, no model or performance claim is made in C1. If those inputs prove too weak, the team should revise the task with the instructor.

Possible dashboard: for an operations planning analyst reviewing where to investigate, show P90, median, eligible n, missing/excluded share and a coverage warning by borough and dispatch hour. Filters should include year, month and geography, comparisons should use compatible definitions and flag partial periods. This follows the EDA's geographic and hourly differences and its denominator/coverage limits. It is a proposed decision aid, not a finished C1 dashboard.

Process record

Decisions and checks

Stage	Decision, alternative and check	Evidence
W1, 24-25 Sep	Compared geography/time and volume/time lines, chose the former provisionally because duration and borough support a bounded diagnostic.	data/W1_selection_evidence.pdf
C1 data	Retained valid zero and long durations, excluded exact copies and conflicting IDs instead of selecting arbitrary versions. Reconciled row flow and independently recomputed P90.	analysis/C1_LFB.ipynb, data/case_tables/audit.json
C1 scope	Kept Unknown geography in global results, excluded incomplete December 2024 from persistence sensitivity. Did not treat absent day as zero demand.	data/case_tables/review_sensitivity.csv, daily_coverage.csv

Individual contributions

Member	Reported C1 contribution and check	Inspect
Matías Muñoz Hoffmann	Proposed the fire service topic, selected London, supplied the assignment instructions and confirmed group 11. Reviewed the row flow counts and the P90 definition/result, checking that 384,627 eligible rows are vehicle mobilisations rather than distinct emergencies.	data/case_tables/row_flow.csv; summary.json; analysis/C1_LFB.ipynb
Clemente Ibarra	Reviewed the borough/hour comparisons and the report's limitations. Checked that Hillingdon and the 11:00 GMT group are leads for further investigation, not proof of poor performance or a causal explanation.	data/case_tables/borough_metrics.csv; hour_metrics.csv; report section 4

AI use and verification

Tool used: OpenAI Codex (technical and coding assistance).

Scope of use: The tool was used as support to explore the technical feasibility of the fire department dataset, assist with code syntax for data quality tests, and structure the initial draft of the C1 deliverables according to the rubric.

Team decisions and criteria:

- The team defined the focus on the London Fire Brigade (LFB) and decided to limit the analysis to the 2023-2024 period to ensure operational relevance and time consistency.
- After evaluating different dispersion metrics, the team chose to use P90 and the median to properly capture variability without distorting the distribution with extreme values.
- Strict methodological rules were established regarding automated suggestions: we rejected artificial imputation of the target variable, avoided arbitrary removal of valid long times, and prohibited using post dispatch variables (such as actual travel or arrival time) to prevent data leakage in future models.
- It was verified that the unit of analysis strictly corresponds to individual vehicle mobilisations and not to single incidents.

Individual reflections confirmed by team representative

Matías: His contribution was selecting the London case and reviewing the row flow and P90. He reports learning to distinguish a vehicle mobilisation from an incident and to interpret P90 without confusing it with an average or an official first-arrival target. His unresolved question is how the observed 20-minute ceiling in the published extract affects geographic comparisons.

Clemente: His contribution was reviewing borough/hour comparisons and the interpretation limits. He reports learning that a higher P90 in Hillingdon identifies a topic for investigation but does not prove worse staff performance or its cause. His unresolved question is how distance and incident type would change the comparisons. These statements were confirmed by Matías on behalf of both members on 29 September 2026; Clemente's direct message was not obtained in this record.

References

- London Fire Brigade / London Datastore. London Fire Brigade Mobilisation Records, 2021-2024 CSV and Mobilisations Metadata. Retrieved 23 September 2026. <https://data.london.gov.uk/dataset/london-fire-brigade-mobilisation-records-24r65>
- Mayor of London (2023, 3 April). MD3051: London Fire Commissioner's Community Risk Management Plan, Your London Fire Brigade 2023-29. <https://www.london.gov.uk/md3051-london-fire-commissioners-community-risk-management-plan-your-london-fire-brigade>
- London Fire Brigade (2024, 5 February). FOIA 8420.1: Response times of fire brigades and data collation. <https://www.london-fire.gov.uk/media/8863/foia84201-response-times-of-fire-brigades-and-data-collation-response.pdf>
- Group 11 (2026). W1: Two London firefighter data lines, provisional selection. Submitted W1 report, included in data/W1_selection_evidence.pdf.
- Universidad del Desarrollo / Tomás Fontecilla Correa (2026). C1: From the W1 project to defensible evidence. Course brief and rubric supplied by the team on 29 September 2026.